

Change History

Application Release Number	Changes Made – Reasons for the Change
b2005	Initial Version

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1. Introduction

What is the System Name, Configuration Item Number and Software Version? Learning Management System (LMS) b2005, CI # 408321	
☐ New Implementation: This is a new system implementation, considered an initial release per IT-OCD-5mS1, <i>IT Standard for Computerized System Validation</i> .	☒ Upgrade to existing implementation: This implementation is an upgrade to the current validation implementation of the system listed below - SuccessFactors LMS
☐ Check this if this implementation replaces an existing system? Provide details on the application being replaced:	Provide application name and current production version being upgraded: SuccessFactors b1808 to b2005
	Based on IT-OCD-5mS1, IT Standard for Computerized System Validation, mention the implementation effort: ☒ Medium Release ☐ Large Release
What are the names of the user groups that this validation effort applies to? People Services, HRIT, R&D, GPS, Compliance & Ethics	
What is the GAMP 5 Classification?	☐ C3 ☒ C4 ☐ C5
What is the Primary Service Model? If "Other" explain:	☒ SaaS ☐ PaaS ☐ IaaS ☐ Internally Hosted ☐ Other
Is the application being developed and/or deployed in phases? If yes, specify: Annual upgrade will include digitization, release b2005 and data migration and will be deployed in 1 phase. SAP is deploying three internal releases for LMS b2005 upgrade – 1H2020p5, 1H2020p16 and 1H2020p26. The release 1H2020p26 will be deployed in production for BMS. Testing will be in 2 phases: Phase 1 of testing includes System Testing and User Acceptance Testing for all requirements (except reports); for reports – only 22 GxP critical reports (Refer Appendix13 for list of reports). System and UAT Testing will be completed against 1H2020p16	☐ Yes ☒ No

Phase 2 includes Regression Testing for patch updates from SAP. Regression testing will be performed by HRIT after SAP deploys 1H2020p26 release, to ensure that system functionalities are not negatively impacted in 1H2020p26. No regression testing will be performed on eSetup and Infinity interfaces. Data migration testing for annual upgrade includes only Phase 1 of OneLMS data migration only. <u>Following not being tested in Annual Upgrade b2005 Release and not in scope for this-</u> 10 reports are being validated in a separate small release; remaining 55 GxP critical reports being validated in regression testing in January 2021; all non-GxP reports will be validated in March-April 2021. (Refer Regulatory Risk Assessment for entire list of Reports).	
Based upon the results of the updated regulatory risk assessment, a validation process must be completed for this system. The validation activities, which are described at a high level in this document, are supported by individual documents, which are considered deliverables for this project. Refer to section 11, <i>Validation Deliverables</i>, for more information. At the conclusion of the validation process, a Validation Summary Report will be generated to describe the process as it occurred and to explain any task level deviations from this Validation Plan.	
Additional Notes: N/A	

2. System Description

LMS is used by BMS training organization worldwide and is the system of record that documents company-wide employee training history. LMS supports the management of training activities such as scheduling, registration, deployment of learning programs, and generation of related metrics. In 2014, the LMS moved from a BMS-hosted system to a Software as a Service (SaaS) solution hosted by SuccessFactors, an SAP Company.

Currently, the system is comprised of the Learning Management System (LMS) as well as the BizX component. BizX provides the Learner data import to the LMS system. BizX accepts the daily feed from the META directory which updates the user record in BizX. BizX then passes the Connector-SF files to the LMS which updates the Learner records in the LMS.

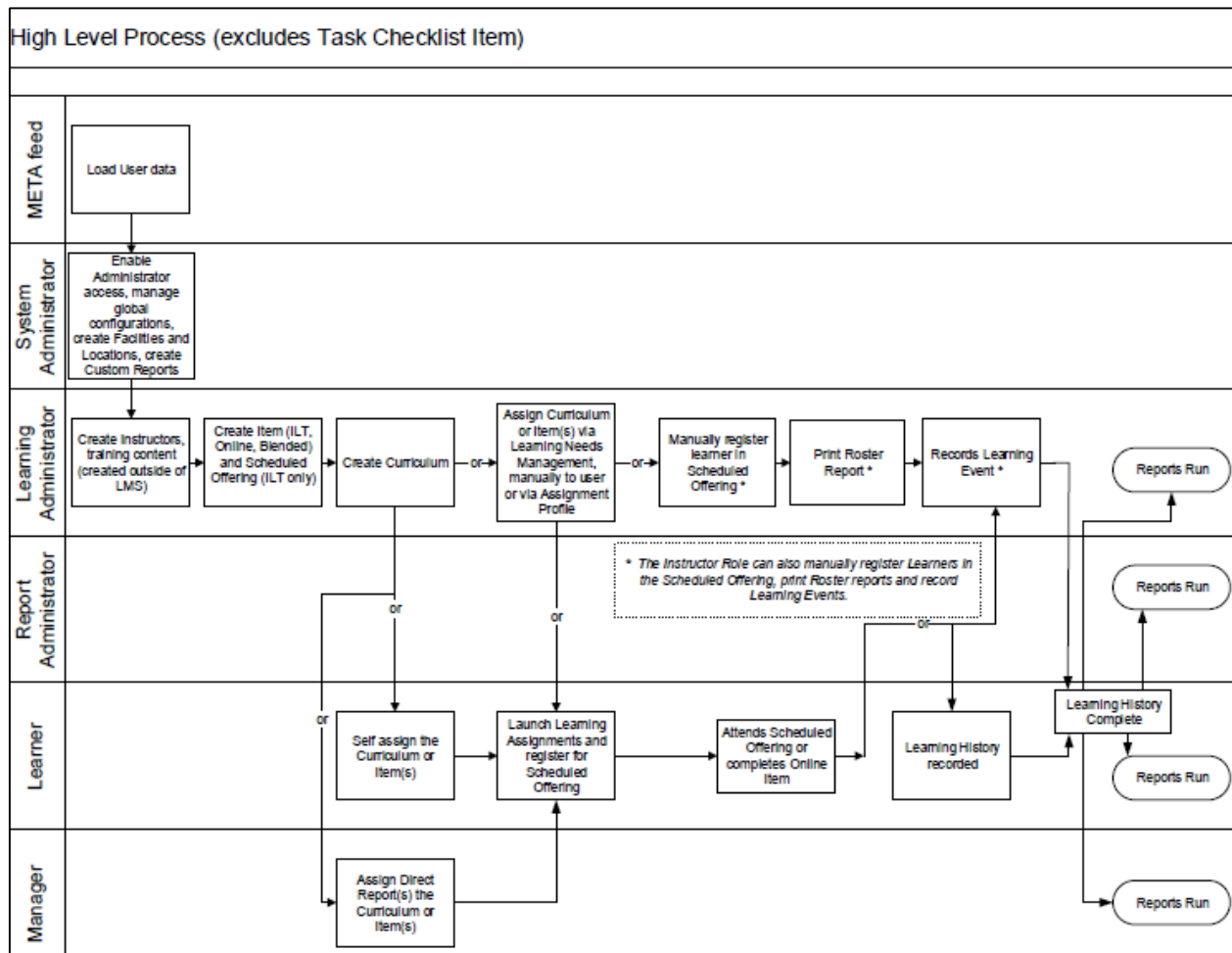
In addition to the Learner data import functionality, BizX provides the user authentication capability. BizX integrates with SAML (Security Assertion Markup Language) for user authentication. High level roles (Everyone (User), System Administrators and Regular Administrators) with corresponding permissions are managed in BizX; to provide entry into LMS after which LMS roles (System Administrator, Service Desk Administrator, Learning Administrator, Reporting Administrator and Restricted Administrator) with corresponding permissions take over and based on the LMS roles; appropriate workflows are enabled. Details on the BizX high-level roles and LMS roles are captured in the LMS Configuration Specification document. Details around roles are captured in the LMS Requirements Specification Table 3-2 Roles and Responsibilities, Section 4.4 User Roles and Access Requirements and Section 4.5 Security Requirements

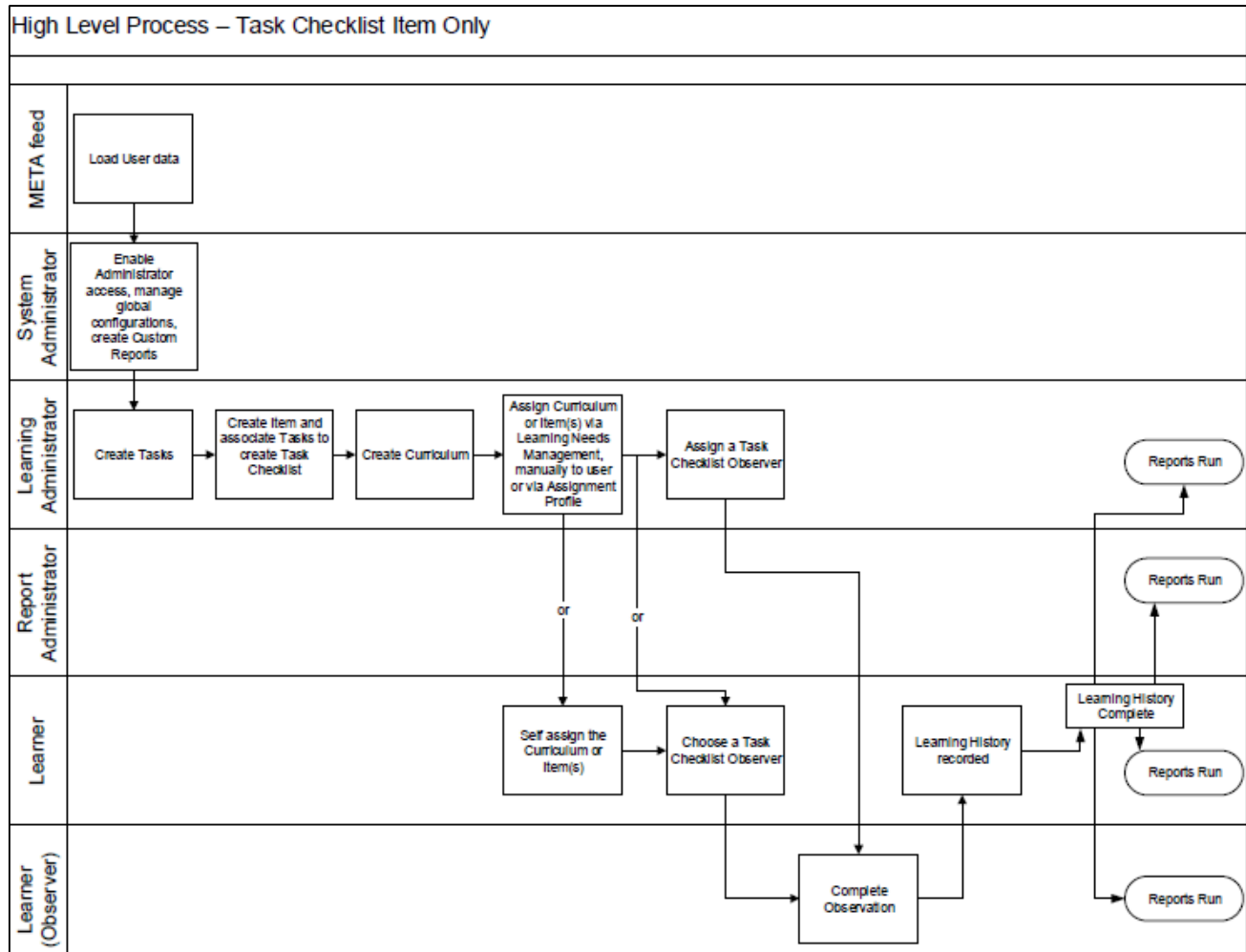
A separate application-specific authentication is being used in the LMS for Administrators who support Learners in countries with data privacy concerns and System Admins with full configuration access. Details around this is captured in LMS Requirements Specification document, section 4.4, User Roles and Access Requirements and section 4.5, Security Requirements.

LMS will be upgraded to the vendor provided version b2005, which is planned to be implemented into the Production environment in December 2020 as part of this release. The system integrates with internal and external content servers to allow Learners to launch training courses.

Below is a high-level process flow diagram.

Figure 2-1 Process Flows





2.1 Product/Supplier Selection

Was this a purchased application (either off the shelf or customized)?		☒ Yes ☐ No
Was a supplier assessment completed?		☒ Yes ☐ No
If Yes, ○ Specify when the supplier assessment completed:	Supplier Assessment Date: SuccessFactors is a SAP company. A re-assessment for SAP Societas Europaea (SE) vendor was conducted in two sessions (Session 1: July 9-11, 2019, Session 2: October 7-11, 2019) as a joint initiative between the Life Sciences Advisory Council (LSAC) and SAP. The supplier audit found the SAP vendor system to be adequate with no major issues. For more details, refer to Governance Risk and Compliance (GRC) record in vitalize - AUD0000896, 2019 SAP Life Sciences Advisory Council (LSAC) Audit.	

☐ Were the results of the assessment successful?		☒ Yes ☐ No
☐ Were any issues identified as part of the assessment that impact the validation strategy being employed for this project?		☐ Yes ☒ No
☐ Summarize what impact the assessment has on the validation strategy and reference the sections of this document that include specific details	There is no impact on the validation strategy due to the supplier assessment, as results of the audit were satisfactory.	
☐ Was negative and boundary testing reviewed during the assessment?	☐ Yes ☒ No Notes: BMS was satisfied with SAP SDLC practices as part of Supplier Assessment.	
If a supplier assessment was not completed, will one be completed as a result of this validation effort?- N/A		☐ Yes ☐ No
☐ If no, provide a justification as to why an assessment is not needed for this release of the application	☐ Justification: N/A	
Additional Notes: N/A		

2.2 Security and Data Integrity

Has a Data Integrity (DI) assessment been completed on this application? ☐ Yes. ☒ No. Select a justification below. ☐ DI controls are built into the CSV templates that will be used for this validation effort. ☒ Other: DI Assessment will be performed during this b2005 upgrade release implementation and result will be captured before VSR.	
How is access to the application controlled? Notes All users (BMS Employees, Consultants and Business Partners) and interfacing applications, regardless of site location access LMS through BMS Single Sign On (SSO) using Security Assertion Markup Language (SAML) authentication with a BMS User ID and password. Additionally, a separate application-specific authentication is being used in the LMS for Administrators who support Learners in countries with data privacy concerns and System Administrators with full configuration access. There are three ways BMS users can login to LMS: Internal users: These users log onto the BMS network using their BMS ID and password and have access to functionality and content as per their role. LMS	☒ Enterprise Active Directory ☐ Access Management Directory ☒ Application Specific ☐ Other:

authenticates them against SAML. When taking training they have access to internal BMS content, BMS internet facing content, third-party content, and content hosted on the LMS servers. - Virtual Private Network (VPN) users: These users log into the BMS network using a VPN connection. LMS authenticates them against SAML. Once they have logged on they have the same rights and functionality as if they logged directly into the BMS network. - Users from Internet: These users log onto the LMS application using their BMS ID and password. LMS authenticates them against SAML. They can only access specific third-party content, and content hosted on the LMS servers.	
Are there specific roles defined for users that restrict what they can and cannot do? Notes: High level roles (Everyone (User), System Administrators and Regular Administrators) LMS roles (System Administrator, Service Desk Administrator, Learning Administrator, Reporting Administrator and Restricted Administrator)	☒ Yes ☐ No
Will third parties or any external users access the application? (E.g. CRO, CMO, etc.) Notes: BMS employees, contractors and external business partners have access to the application.	☒ Yes ☐ No
Does the system have an audit trail that tracks all required information for all GxP records? Notes: N/A	☒ Yes ☐ No
Is the audit trail in compliance with Electronic Records/Electronic Signature regulations and Data Integrity guidance? Notes: E-signature not applicable (as captured in RRA). DI assessment is in progress and any gaps for audit trail will be captured in VSR	☒ Yes ☐ No
If this is an existing application, are there any changes being made to the current security model as part of this release? Notes: N/A	☐ Yes ☒ No
Additional Notes: N/A	

2.3 External Interfaces

Does this application have external interfaces including tools, applications, or components?		☒ Yes ☐ No
If yes, where are these interfaces documented?	☒ External document: LMS Architecture Document (QSD- 000076)	

	☐ This document
If this is an existing application, are there any changes being made to external interfaces as part of this release? Notes: N/A	☐ Yes ☒ No
Were the business owner and CI owner of the interface notified if there are changes to external interfaces? Notes: N/A	☐ Yes ☐ No ☒ N/A
Additional Notes: N/A	

Hardware and Software Components/Tools

What is the communication architecture model of the application? Notes: LMS is a SaaS (Software as a Service) solution.		☐ Client/Server ☐ Web Based ☒ Cloud Based ☐ Other
Where is the hardware and software infrastructure located?	☐ Hopewell Data Center ☐ Lawrenceville Data Center ☐ Other BMS Data Center (e.g. Wynyard Data Center, UK) ☒ SaaS Provider ☐ Amazon Web Services (AWS) Notes: N/A	
How do/will the users communicate with the application?	☐ Through existing intra-site data communications infrastructure ☒ SaaS ☐ Other Notes: See Section 2.2 Security Data Integrity above.	
Where are the details for the hardware and software components for this application documented?	☒ External document ☐ This document Note: Application version is b2005.16. SAP SuccessFactors is a cloud based application, so information for Hardware and software configuration is not available.	
If this is an existing application, are any changes being made to hardware/software configuration as part of this release? Notes: N/A		☐ Yes ☒ No
Additional Notes: N/A		

3. Regulatory Risk Assessment

Is a Regulatory Risk Assessment required for this release of the application?	
☒ Yes The regulatory risk assessment was updated according to IT-OCD-5mS1, <i>IT Standard for Computerized System Validation</i> . The following summarizes the results of the risk assessment process completed on this application. See Regulatory Risk Assessment for the application for more details.	☐ No The regulatory risk assessment was reviewed and an update is not required for this release.

Overall risk category assigned to this application?	☐ High ☒ Medium ☐ Low ☐ None Notes: N/A	
What is the GAMP category assigned to this system as a result of the Regulatory Risk Assessment?	☐ C3 (non-configured software) ☒ C4 (configured software) ☐ C5 (custom software)	
With which of these requirements does this application need to comply?	☒ Electronic Records ☐ Electronic Signatures ☒ Open System ☐ Hybrid System	
If this is an upgrade to an existing application, does the change impact the regulatory risk assigned to the system? Notes: N/A	☐ Yes ☒ No	
Additional Notes: N/A		

4. Validation Scope

The following sections define the specific items that are in scope and out of scope for this validation effort.

In Scope	
What is the scope of the validation effort?	☒ <i>Whole System</i> ☐ <i>Partial System</i> Notes: 1) b2005 implementation is an annual upgrade of SuccessFactors. The following functionality is being updated as part of the annual upgrade- • New Learning Administration User Interface will be implemented SAP redesigned the Learning Administration user interface to enhance the administrator's experience • Custom field at an item to reflect GxP related information- Added a custom field to item to provide an information if item is GxP related or non-GXP item. • New user fields in META view and Enterprise Look-up Tool- Added new user field in META for Commercial, GPS and R&D. This will

	allow GPS & R&D to assign assignment profiles depending on the user's State, Job Title. • Content level auditing report update- Update content level auditing report to make it more user friendly. Reformat the current data to make it reader friendly. Removing existing not frequently used columns & keeping the frequently used columns. • Enabling Quiz/Assessment feature- Allow Learning Admin to create assessment within SF, instead of Pedagogue • Curriculum point in time report update- Create Curriculum point in time report. Allow admins to view what sub-curriculums/items were part of parent curriculum during a specified timeframe. • Update training notifications for Learners/Managers- Review/update the notifications as a part of the upgrade. 2) All the Requirements, Configuration/Functional documents and Test Scripts are being moved to VERA ALM from DocIT as part of digitization effort. 3) Also, Phase 1 of data migration of OneLMS project (OneLMS project is to migrate LMS data from Two Applications, Celgene & Juno Compliance Wire to SuccessFactors). More details will be captured in the OneLMS Data Migration Plan 4) 22 GxP critical reports will be fully validated as part of this Release. All other reports will be validated via other change requests (not included in this annual upgrade). See Appendix 13 for list of the reports.
What are the validation activities that are in scope of this effort?	☒ Data migrations ☐ Infrastructure qualification ☒ Installation qualification ☒ ERES and Data Integrity controls ☐ Interfaces to other BMS applications ☒ Interfaces to third party applications

	☐ Retirement of application being replaced ☒ Testing/verification of items listed in the scope of the validation effort ☐ Other Notes: SAP SuccessFactors will perform AplQs in accordance with their procedures. SAP SuccessFactors will provide BMS with documented evidences of successful AplQ for the test and production environments for LMS version b2005. A separate AplQ Plan will not be developed for this SaaS solution. The Validation Summary Report will capture the installation results.
Is this a local/global system?	☐ Local system ☒ Global system ☐ Other: Notes: N/A
Which business units will be involved in this validation effort?	Business Unit(s): People Services
Is the system release being deployed in phases?	Annual upgrade will include digitization, release b2005 and data migration and will be deployed in 1 phase. SAP is deploying three internal releases for LMS b2005 upgrade – 1H2020p5, 1H2020p16 and 1H2020p26. The release 1H2020p26 will be deployed in production for BMS. Testing will be in 2 phases - Phase 1 of testing includes System Testing and User Acceptance Testing for all requirements (except reports); for reports – only 22 GxP critical reports (Refer Appendix13 for list of reports). System and UAT Testing will be completed against 1H2020p16 Phase 2 includes Regression Testing for patch updates from SAP. Regression testing will be performed by HRIT after SAP deploys 1H2020p26 release, to ensure that system functionalities are not negatively impacted in 1H2020p26. No regression testing will be performed on eSetup and Infinity interfaces. Data migration testing for annual upgrade includes only Phase 1 of OneLMS data migration only.

Out of Scope	
Are any specific items considered out of scope that were not explained above?	☒ Yes ☐ No Notes:
Additional Notes: The items that are out-of-scope for this validation effort include: • Development and testing of the core product – SAP SuccessFactors is responsible for developing and testing their core commercial product according to their procedures including requirements, design, and other documentation. • 10 reports (GxP critical) which will be tested as part of the small release. • Remaining 55 GxP critical reports will be tested in regression testing as part of January patch release. • 99 Non-GxP reports will be tested in April Patch Release. • Also Phase 2 and Phase 3 of OneLMS data migration are out of scope. • Infrastructure qualification – SuccessFactors is responsible for qualifying the supporting infrastructure according to their procedures. • Archiving – All data (including metadata) is planned to be maintained in the active database. Therefore, a data archival strategy is outside the scope of this validation effort.	

5. Roles and Responsibilities

Are the roles and responsibilities for this validation effort outlined in another document? ☐ Yes ☒ No If yes, specify the document here: The table below defines the responsibilities for each functional role that will be involved in the validation process along with the individual(s) assigned to that role. For a complete list of documents that each role is responsible for approving, see Table 11-1 Validation Deliverables. For the general responsibilities associated with each of these roles within the validation process, see Table 5-1 Roles And Responsibilities in BMS-SOP-5m, <i>Computerized System Validation</i> and IT-OCD-5mS1, <i>IT Standard for Computerized System Validation</i> .

Table 5-1 Roles and Responsibilities

Role	Resource(s)/Organization	Project Specific Responsibilities
Business Owner	Cathy Twarog Global Learning Technology Specialist - People Services – Talent Management	• Participates in regulatory risk and supplier assessments. • Manages UAT effort. • Complete training on CSV procedures.

Role	Resource(s)/Organization	Project Specific Responsibilities
Digital Capability Manager/Configuration Item Owner	Khalid Deshnoor Manager , Learning -Digital Capability Management - HRIT	- Participates in risk assessment and supplier assessments. - Leads and ensures the documentation of the selection, design, acquisition, development, unit testing, configuration and implementation of technology solutions. - Manages the process for specification and acquisition of necessary hardware and software infrastructure. - Manages System Testing effort.
Project Manager	Shiv Kumar Lund Consultant – Enabling Functions	- Manages business application system implementation effort. - Provides business and technical input during the validation process.
Validation Lead	Brijesh Setia Manager – Computer System Validation(CSV), IT Quality Management Chandra Immareddy Consultant – CSV, IT Quality Management	- Ensures the validation process is conducted per IT-OCD-5mS1, <i>IT Standard for Computerized System Validation</i>, providing guidance on validation strategy, tasks and deliverables. - Ensures that all non-Information Technology staff members have completed training on IT-OCD-5mS1, <i>IT Standard for Computerized System Validation</i>. - Ensures that validation documentation is maintained, controlled, and available for regulatory review on behalf of the Business Owner.
Technical Lead	Kevin Weatherbe Consultant - HRIT	Provides technical expertise and documentation of the selection, design, acquisition, development, unit testing, configuration and implementation of technology solutions.
Test Lead	Vasu Shirish Karadage Consultant – Enabling Functions	Manages system testing and UAT testing.
Installation Support (Infrastructure)	SAP SuccessFactors Product Team	Installs and verifies the correct operation of the hardware and software infrastructure.
Installation Support (Application)	HRIT Team	Perform application installation activities
Regulatory Compliance Subject Matter Expert (SME)	Kameswari Yedlapalli Analyst, GxP IT Quality Assurance and E-Compliance	Provides regulatory guidance and oversight to the project team.

6. Development

The following sections explain the activities that will occur during the development of this system. The below sections include analysis, design, build/configure, and unit/integration testing.

Is this a vendor supplied application? (COTS, SaaS)

☒ Yes (*Select all that apply*)

☐ No

☐ Vendor documentation will not be leveraged for this validation effort and the development team will follow IT-SOP-1, *Solution Lifecycle*.

☐ Application is developed in-house. The development team will follow IT-SOP-1, *Solution Lifecycle*.

☒ Based on the results of the supplier assessment and regulatory risk assessment, BMS is planning to leverage supplier documentation for this validation effort, as these have shown that the vendor is in compliance with BMS requirements as outlined in IT-SOP-1, *Solution Lifecycle*.

Which of the following vendor supplied documents will be utilized for development activities?

- ☐ Design specification
☐ Configuration specification
☒ Application installation scripts
☐ Unit and Integration testing
☐ Other.

Additional Notes: N/A

6.1 Analysis

The Requirements Specification defines the user's expectations for the production computerized system including business, compliance, quality, and technical needs.

What is the status of the Requirements Document?

- ☐ Will be created
☐ Has been created and approved
☒ Will be updated for this validation effort

Notes: LMS Requirements Specifications will be stored and approved in ALM

Where will requirements be captured?

- ☒ A living requirements specification document
☐ A project-specific requirements specification document

	Note: MS Requirements Specifications will be stored and approved in ALM as part of digitization effort for SuccessFactors Learning Management System A report will be generated for the User requirement Specification and uploaded as reference document in DocIT to baseline the requirements for b2005 release
Which roles/business units were involved in gathering the requirements?	IT/Business Units and Roles involved in gathering requirements: BMS Business Owners, Technical Representatives, ITQM Lead and the Project Manager were involved in gathering the requirements specifications.

6.2 Design

Which of the following type of design document will be updated as a result of this release?	
☐ Provided by the vendor ☐ Functional Design/Functional Requirements ☐ System Design ☐ Detailed Design/Technical Design ☐ Architecture ☐ Database Design ☐ Other Notes:	☒ By BMS IT personnel ☒ Functional Design/Functional Requirements ☐ System Design ☐ Detailed Design/Technical Design ☐ Architecture ☐ Database Design ☐ Other Notes: Functional design document will be updated to reflect changes/enhancements for this release in ALM.
Are there multiples of the document type that were selected in the previous row? ☐ Yes ☒ No	
Additional Notes: N/A	

6.3 Build/Configure

What build/configuration activities are required for this release?	
☒ Configuration ☐ Custom Code ☐ Other Notes: N/A	
What tool will be used to manage and control the source code/configurations for this application?	☐ CollabNet TeamForge (CI#389912) ☒ Not Applicable ☐ Other

	Notes: This is a Software as a Service (SaaS) solution and hence all code development is the responsibility of the vendor. All custom configuration changes for the custom reports are applied by the BMS team through the application's front end for the custom reports. All other custom configuration changes are managed and controlled through the execution of AplQ scripts during the respected release. There are no backend configuration files applied on the application server by BMS.
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Is the source code repository qualified?	☒ Yes ☐ No If no, justify: SAP SuccessFactors base code is managed by SAP Vendor in accordance with their documented procedures
What document will be used to capture the configuration details of this system?	
☐ Provided by the vendor ☐ Configuration Specification ☐ IQ documentation ☐ Other Notes: N/A	☒ By BMS IT personnel ☒ Configuration Specification ☐ IQ documentation ☐ Other Notes: Configuration Specification Document will be updated and uploaded to ALM.
Additional Notes: N/A	

6.4 Unit Testing

Which of the following sections indicates the Unit Testing for this release?		
☐ C3 Non Configured System ☐ Not Required ☐ BMS will only complete negative and boundary testing	☒ C4 Configured System ☒ Performed by the supplier ☐ Performed by the BMS development team ☐ Not to be performed Notes: N/A	☐ C5 Custom System ☐ Performed by the supplier ☐ Performed by the BMS development team ☐ Not to be performed Notes: N/A
How will Unit Testing be documented and where will the documentation be stored?	☐ Via checklists that will be stored in <EDMS/DocIT> ☐ Via scripts that will be stored in <ALM> ☐ Other Notes: N/A	
Will Unit Testing include negative and boundary testing? ☒ Yes ☐ No Notes: SuccessFactors is a SaaS system, all unit testing is performed by the vendor. Unit testing for custom code will be performed by BMS		
Additional Notes: N/A		

7. Testing

The following sections explain the activities that will occur to ensure the system is properly installed into one or more test environments and formally tested.

7.1 Infrastructure Installation Qualification/Operational Qualification

Infrastructure Installation Qualification/Operational Qualification (IQ/OQ) provides assurance that the supporting infrastructure is installed and configured properly prior to testing or production use.	
Who will complete Infrastructure qualification activities?	☐ DXC according to their procedures ☐ BMS Infrastructure Services according to their procedures ☒ Other: SAP SuccessFactors ☐ Not Applicable (because IQ is not in scope)
Additional Notes: SAP SuccessFactors is responsible for conducting the Infrastructure Qualification in accordance with their procedures.	

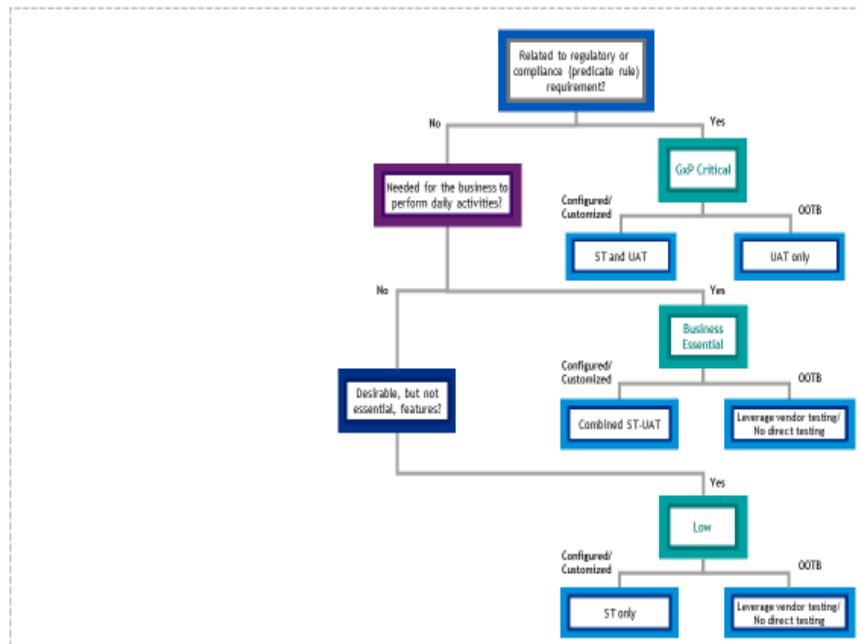
7.2 Application Installation Qualification/Operational Qualification

IQ provides assurance that the application is installed and configured properly prior to testing or production use.	
Will there be a separate IQ plan and summary for this validation effort?	
☒ No (Select all that apply)	☐ Yes (Select all that apply)
At which of the following points will IQ be executed? ☒ Prior to System Testing ☐ Prior to UAT ☒ Prior to application's release into production. Notes: N/A	☐ An IQ Plan will be created and approved prior to the execution of individual IQ scripts and/or procedures. The IQ Plan will document the environment description, hardware and software components, roles and responsibilities, supporting technical documentation and procedures, and expected deliverables for the IQ documentation. ☐ An IQ Summary Report will be created after the system goes into production. The IQ Summary Report will summarize the IQ process and results for all environments and any deviations from the IQ Plan.

IQ execution documents the technical specification of the component being installed, the installation procedure, verification steps, pass/fail indicator and verifiable objective evidence. How will IQ execution be documented? ☒ Test script executions in ALM (for configuration completed by BMS) ☐ Test script executions on paper. These scripts will be stored: ☐ Work Instructions ☒ Other: The vendor SAP SuccessFactors will provide a summary report of the AplQ which will be attached to the vitalize change request.	
Who will be performing IQ? SAP SuccessFactors will perform the AplQ in accordance with their procedures. SAP SuccessFactors will provide BMS with documented evidence of successful AplQ for the test and production environments for LMS version b2005. A separate AplQ Plan will not be developed for this SaaS solution. The Validation Summary Report will capture the installation results. Notes: The vendor documents the installation in the Operations Automated Deployment Report which will be attached to the project's Change Request. BMS completes IQ scripts for configuration updates and this is captured in Validation Summary Report.	

7.3 Testing Strategy

Per RRA, requirements were divided into GxP Critical, Business Essential and Optional requirements. Recommended validation strategy is as below-



Any exceptions will be documented in the Test Plan with appropriate justification.

What type of testing will be completed for this validation effort?

- ☒ System Testing (ST)
☒ User Acceptance Testing (UAT)
☐ Combined ST/UAT
☒ Data Migration Testing (DM)

7.3.1 System Testing

Will there be a separate plan and summary for ST?

Test plan documents test procedures, test environment, and the areas to be tested. This plan is approved prior to the start of ST activities. Documented test scripts drive execution of system testing.

☒ Yes

☐ A System Test Plan will be created. Once system testing is completed, a summary report will be generated to document the testing process and findings. The System Test Summary Report will be approved prior to the start of user acceptance testing execution activities.

☒ A combined ST/UAT plan will be created. Once system testing is completed, an ST Closure Memo will be generated and approved prior to the start of UAT execution. A combined ST/UAT Summary report will be generated to document testing process and findings after completion of UAT.

☐ Other:

☐ No

☐ This product was developed by a supplier. The SDLC process utilized by the supplier was reviewed during the supplier quality assessment and determined to be acceptable.

	☐ The configuration of this application is being confirmed as part of the IQ process. Therefore, a separate ST plan is not required. ☐ Other: Notes: N/A
Who will execute the System Test?	System Testing Role/Group Name: App Support ACN SuccessFactors IT
How will ST execution be documented?	☒ Test script executions in ALM ☐ Test script executions on paper. Paper test scripts will be stored in: ☐ Other:
Additional Notes: BMS is relying on the vendor to perform regression testing. During previous releases the regression testing did not encounter any issues and the system is mature. Since implementing b1308 the base functionality has not changed. ST and UAT Testing will be completed in Phase 1 of Testing for functionality updates & reports Regression testing will be done in Phase2 for any patch updates from SAP	

7.3.2 User Acceptance Testing

Will there be a separate plan and summary for UAT? Test plan documents test procedures, test environment, and the areas to be tested. This plan is approved prior to the start of UAT activities. Documented test scripts drive execution of user acceptance testing.	☒ Yes ☐ A UAT Plan will be created and approved prior to start of UAT execution. Once UAT is completed, a summary report will be generated to document the testing process and findings. UAT Summary report will be approved prior to production implementation. If installation for the production environment starts prior to the completion of UAT, a promote to production memo must be completed. ☒ A combined ST/UAT plan will be created. Once system testing is completed, an ST Closure Memo will be generated and approved prior to the start of UAT execution. A combined ST/UAT Summary report will be generated to document testing process and findings after completion of UAT. ☐ No ☐ The validation effort for this project is simple and straightforward and as a result, the project team determined that the content of the User Acceptance Test Plan would be included in this validation plan. ☐ Other Notes: N/A
Who will UAT be executed by?	User Acceptance Testing Role/Group Name: Research and Development, Global Product Development & Supply and Commercial
Will impacted business units be engaged in the UAT process?	☒ Yes ☐ No

How will UAT execution be documented?	☒ Test script executions in ALM ☐ Test script executions on paper. Paper test scripts will be stored in: ☐ Work instructions ☐ Other: Provide details- N/A
Additional Notes: N/A	

7.3.3 Data Migration

Is data migration included in this validation effort?		☒ Yes ☐ No
Which of these deliverables will apply?	☒ Data Migration Plan ☒ Data Migration Testing/Verification ☒ Data Migration Summary for Test ☒ Data Migration Summary for Production ☐ Data Migration Summary (Combined for Test and Production)	
Will there be a separate plan and summary for data migration testing? Data Migration Plan documents the migration process including planning, scope, development and verification activities, and acceptance criteria.		☒ Yes ☐ No
Additional Notes: Data migration in annual upgrade only includes Phase1 of OneLMS migration. Data migration summary reports for annual upgrade will only include Phase 1 of OneLMS migration.		

7.3.4 Traceability

Is the traceability of the system being updated or changed as a result of this validation effort? Notes: Traceability Matrix will be generated from ALM and will be a snapshot in time.		☒ Yes ☐ No
How is the traceability matrix being managed for this validation effort?	☐ Project Specific Document ☒ Living Document ☐ Combination of Project Specific and Living Document	
What is included in the traceability matrix?	☐ Requirements to Design ☒ Design to ST ☒ Requirements to UAT ☐ Requirements to ST/UAT ☐ Scope Items/Bug Fixes to ST/UAT ☐ Other: N/A	
Additional Notes: <i>Traceability Matrix will be extracted as a report after the testing and approved in DocIT. Any untested reporting requirement (as part of this annual upgrade) will be mapped to the old releases.</i>		

8. Deployment

To which business groups will the system be deployed?	Business Groups: Enabling Functions (Corporate), Research and Development, Global Product Development & Supply and Commercial
Will a separate deployment plan be created for this validation effort?	☐ Yes. ☒ No
What deployment activities will be completed for this validation effort?	☒ A change ticket will be created and approved. ☐ Applicable training activities will be completed. ☒ IQ will be executed to document the installation and configuration in the production environment. ☒ Validation Summary Report (VSR) will be created to summarize the effort and document any deviations (variation from this plan or governing procedures) that occurred. Promote to Production authorization will be ☒ Documented in the VSR ☐ Documented in a separate promote to production memo. Release for use authorization will be ☒ Documented in the VSR ☐ Documented in a separate release for use memo
Additional Notes: N/A	

9. Supporting Procedures and Documentation

9.1 Production Support

Will BMS Service Desk be the initial contact for application support?	☐ Yes ☒ No. If no, specify. See Additional Support section below. ☐ To be determined. The production support information will be documented in the Validation Summary Report.
Who will provide additional support as required?	Additional Support: SuccessFactors users contact People Services first, unresolved questions and issues are escalated to the BMS Service Desk. Any unresolved questions or issues are furthermore escalated to Accenture who will provide third level support. SuccessFactors Administrators contact the BMS Service Desk directly. Unresolved questions and issues are escalated to Accenture who will provide second level support. SAP SuccessFactors will provide support as required and is also responsible for system-level administration activities, which are completed in compliance with their internal procedures.
Who will manage the system administration of the application?	

☒ Accenture is responsible for application system administration activities that are completed in compliance with supplier's application administration processes. All support requests are tracked and resolved in compliance with BMS processes. ☐ <Name of Supplier> is responsible for application system administration activities which are completed in compliance with supplier's application administration processes. All support requests are tracked and resolved in compliance with the supplier's processes.
Who will manage the infrastructure components? ☐ Infrastructure components <will be/continue to be> managed by <IT Infrastructure Services> in accordance with existing procedures. ☐ Application and infrastructure support <will be/is/continue to be> provided by the <list name of supplier>. ☒ Other. This is a SaaS solution, Infrastructure components are managed by the vendor
Will the standard BMS IT support process be utilized? ☒ Yes ☐ No.
Additional Notes: N/A

9.2 Disaster Recovery/System Unavailability

Is this application used in support of a mission or business critical process?	☐ Yes ☒ No This system is not a mission critical application and hence per IT-SOP-2, <i>Infrastructure Lifecycle</i> , BIA and DR plan are not required. ☐ To be determined. The disaster recovery and system unavailability information will be documented in the Validation Summary Report.
Is this application listed on a BIA?	☒ Yes. Specify BIA: This is not required, however a BIA was created. Refer to SuccessFactors LMS Business Impact Analysis QSD-043760. ☐ No.
If the application is listed on a BIA, does it require disaster recovery?	☒ Yes. For detailed information about RTO and RPO information for this application, refer to the BIA referenced above. ☐ No
System unavailability processes are documented in the business procedures. Refer to BIA QSD-043760.	
Additional Notes: N/A	

9.3 Technical Procedures

Select the applicable statement:

- ☐ This is a new application. Procedures to ensure that the application functions correctly and reliably in a production setting are required. The <project team/System Owner> will identify the necessary procedural requirements and develop/update these procedures or use existing ones. A list of the new/updated procedures will be included in the VSR
- ☐ This is an existing application and existing procedures will be followed by Information Technology staff to ensure that the system continues to function correctly and reliably in a production setting. Applicable procedures will be documented in the VSR.
- ☒ This is a Software as a Service application. Procedures to ensure that the application functions correctly and reliably in a production setting are required. These procedures are developed and maintained by SuccessFactors in compliance with their processes.

Additional Notes: N/A

9.4 Business Procedures and Documentation

Are new business procedures/documentation or changes to existing business procedures/documentation required for this validation effort?

- ☐ Yes. The Business Owner will identify the necessary business procedures/documentation and develop/update these procedures/documentation or use existing ones. A list of the new/updated procedures will be included in the VSR.
- ☒ No. No business procedures/policies maintained for SuccessFactors. Individual team maintains their own procedures & work instructions.

Additional Notes: N/A

9.5 Training

Training is required for all users and support personnel.	
Verify that the required approvers of this document completed training on IT-OCD-5mS1, IT Standard for Computerized System Validation.	☒ Training Completed
What type of training is required for this validation effort?	☐ System specific training ☐ Regulatory training (such as electronic records, electronic signatures, or other regulatory guidance documentation). ☒ To be determined. This upgrade is a complete redesign of the administrator interface in SuccessFactors; the details will be captured in the VSR. ☐ None.
If yes, will a separate training plan be created?	☒ Yes. A detailed training plan will be developed by People Services. This plan will cover the required training, essential content, description of how the content will be developed, methods of delivery, identification of delivery resources, and timelines. Because this upgrade impacts the existing business processes, training will include both system functionality and the surrounding business processes. ☐ No. Training content will be developed by <group name> and will include <summarize any additional information, if available>. Training will be <provide information on the delivery methods -- web-based, classroom, train the trainer>.
N/A	
Additional Notes: N/A	

10. Validation Acceptance Criteria

This system will be considered validated and acceptable for production use when the following criteria have been met:

- The system performs the required business functions and is acceptable to the Business Owner and System Owner.
- The system is developed according to the applicable BMS policies and procedures.
- The system meets applicable regulatory requirements.
- All activities and deliverables as required per the Validation Plan have been completed and/or approved deviations have been filed.

11. Validation Deliverables

Each document that is listed as a deliverable in the table below will be sent for review and approval and must align with IT-OCD-5mS1, *IT Standard for Computerized System Validation*.

The table below lists the deliverables that will be produced during the validation of this system and the approvals for each.

The validation documentation for this system will be managed electronically using the below validated system(s):

- DocIT– electronic document repository used for the management of controlled quality system documents.
- Application Lifecycle Management (ALM) – paperless testing tool used for management and documentation of qualification and testing activities.

Table 11-1 Validation Deliverables

Deliverable	Approved By	Planned Storage Location
Regulatory Risk Assessment	Business Owner Regulatory Compliance SME Validation Lead	DocIT
Validation Plan	Business Owner Digital Capability Manager (DCM)/Configuration Item (CI) Owner Validation Lead Regulatory Compliance SME	DocIT
Data Migration Plan	Business Owner DCM/CI Owner Validation Lead Regulatory Compliance SME	DocIT
Requirement Specification	Business Owner DCM/CI Owner Validation Lead Regulatory Compliance SME	ALM
Design Specification	DCM/CI Owner	ALM
Test IQ Scripts (includes system and user acceptance environment scripts)	DCM/CI Owner Validation Lead	ALM
System and User Acceptance Test Plan	Business Owner DCM/CI Owner Validation Lead	DocIT
System Test Scripts (executed including all evidence gathered as a result of testing)	DCM/CI Owner Validation Lead	ALM
System Test Closure Memo	Business Owner DCM/CI Owner Validation Lead	DocIT
User Acceptance Test Scripts (executed including all evidence gathered as a result of testing)	Business Owner DCM/CI Owner Validation Lead	ALM

Deliverable	Approved By	Planned Storage Location
System and User Acceptance Test Summary Report	Business Owner DCM/CI Owner Validation Lead	DocIT
Traceability Matrix	Business Owner DCM/CI Owner Validation Lead	DocIT
Data Migration Testing	DCM/CI Owner Validation Lead	ALM
Data Migration Verification	Business Owner Validation Lead	ALM
Test Data Migration Summary Report- Phase 1	Business Owner DCM/CI Owner Validation Lead Regulatory Compliance SME	DocIT
Requirement Specification ALM Report	Reference Document	DocIT
Validation Summary Report	Business Owner DCM/CI Owner Validation Lead Regulatory Compliance SME	DocIT
Production IQ Scripts	DCM/CI Owner Validation Lead	ALM
Training Plan	Business Owner DCM/CI Owner	DocIT
Production Data Migration Scripts	DCM/CI Owner Validation Lead	ALM
Production Data Migration Verification	Business Owner Validation Lead	ALM
Production Data Migration Summary	Business Owner DCM/CI Owner Validation Lead Regulatory Compliance SME	DocIT

12. Referenced Documents

The documents below are directly referenced in this plan and/or have been relied upon to create this plan.

Table 12-1 Referenced Documents

BMS Procedures/Guidance
BMS-SOP-5m, <i>Computerized System Validation</i>
IT-OCD-5mS1, <i>IT Standard for Computerized System Validation</i>
IT-SOP-1, <i>Solution Lifecycle</i>
IT-SOP-2, <i>Infrastructure Lifecycle</i>

IT-OCD-1G1 (008895), <i>IT Testing Manual</i>
Learning Management System Specific Documentation
LMS Risk Assessment (QSD-038430) in DocIT
Requirement Specification in ALM (Corporate\SuccessFactors\Requirements)
LMS b2005 Configuration Design documents in ALM (Corporate\SuccessFactors\ Configuration Design)
LMS b2005 Functional Design Documents in ALM (Corporate\SuccessFactors\ Functional Design)
LMS b2005 System Architecture (QSD- 000076) in DocIT
SuccessFactors LMS Business Impact Analysis (QSD-043760) in DocIT

13. Appendices

13.1 List of Reports

S.No.	Report Name	Type	Report Format (Output)
1	User_Management-Item_Status_Basic_CSV	Custom	CSV
2	User_Management-Item_Status_Basic_GroupByUsers	Custom	HTML & PDF
3	Curriculum-Assignment_By_Curriculum	Custom	HTML & PDF
4	User_Management-Curriculum_Not_Assigned_CSV	Custom	CSV
5	User_Management-Overdue_Item_Report_Date_GroupByEmpStatus	Custom	HTML & PDF
6	User_Management-Overdue_Item_Report_Date_GroupByEmpType	Custom	HTML & PDF
7	User_Management-Overdue_Item_Report_Date_GroupByItem	Custom	HTML & PDF
8	User_Management-Overdue_Item_Report_Date_GroupByJobLocation	Custom	HTML & PDF
9	User_Management-Overdue_Item_Report_Date_GroupByJobPosition	Custom	HTML & PDF
10	User_Management-Overdue_Item_Report_Date_GroupByOrganization	Custom	HTML & PDF
11	User_Management-Overdue_Item_Report_Date_GroupBySupervisor	Custom	HTML & PDF
12	User_Management-Assignment_Profile_Detail_CSV	Custom	CSV
13	Learning-Curriculum_Percent_Complete_CSV	Custom	CSV
14	Audit-Curriculum_Item_History_CSV	Custom	CSV
15	Audit-User_Custom_Column_History_CSV	Custom	CSV
16	Audit-User_Item_Assignment_Basic_CSV	Custom	CSV
17	User Learning Needs By Employee Status(Sec)	Custom	HTML & PDF
18	User Learning Needs by Employee Type(Sec)	Custom	HTML & PDF

S.No.	Report Name	Type	Report Format (Output)
19	User Learning Needs by Job Location(Sec)	Custom	HTML & PDF
20	User Learning Needs by Job Position(Sec)	Custom	HTML & PDF
21	User Learning Needs By Organization(Sec)	Custom	HTML & PDF
22	User Learning Needs By Supervisor(Sec)	Custom	HTML & PDF

14. Approvals

APPROVALS The electronic signatures of the individual(s) listed below indicate their approval for solution development.	
Role	Participant
Business Owner	Cathy Twarog
Digital Capability Manager / Configuration Item Owner	Khalid Deshnoor
Validation Lead	Brijesh Setia
Regulatory Compliance SME	Kameswari Yedlapalli